JIGYANSU TRIPATHI



Water Resources Engineer | Hydraulic & Hydrological Modelling | Python & GIS

9424636453 jigyansu1703.24@kgpian.iitkgp.ac.in

linkedin.com/in/jigyansu-tripathi-390004181 jigyansu1703.github.io/Portfolio

Bhopal, Madhya Pradesh

SUMMARY

Water Resources Engineer with an M.Tech in Water Engineering and Management from IIT Kharagpur (May 2026), working as a Design Engineer on water supply, wastewater and storm water projects in Madhya Pradesh. Experienced in preparing STP rehabilitation DPRs and cost estimates, and in storm water drainage modelling using SewerGEMS. Skilled in hydraulic modelling of water distribution networks (EPANET, WaterGEMS), GIS/DEM-based spatial analysis and Python-driven data analysis. Strong interest in water infrastructure planning and sustainable water and wastewater systems.

EXPERIENCE

Design Engineer

Vastushilpi Project and Consultancy Pvt. Ltd

05/2026 - Present Bhopal, Madhya Pradesh

Consultancy delivering water supply, wastewater treatment, storm water drainage and flood modelling projects

- Prepare DPRs and cost estimates for STP rehabilitation in 5 towns, and propose new STPs for 6 towns in Madhya Pradesh.
- Develop storm water drainage models in SewerGEMS for Hiraundi Industrial Area, Madhya Pradesh, under the BHAVYA scheme.
- Support water supply and water treatment plant projects in Madhya Pradesh under the Urban Challenge Fund (UCF).

Research Intern

Haldia Water Services Pvt. Ltd

05/2025 - 06/2025 Geonkhali, WB

Company specializing in water treatment and management services

- Conducted performance evaluation of the Geonkhali WTP using four years of operational water quality data.
- Improved water quality analysis efficiency by using an interactive Tableau/Python-based dashboard visualizing seasonal and tidal impacts.

PROJECTS

Hydraulic Simulation and Pressure Management Framework for Water Distribution Networks Using Python and EPANET

01/2025 - 05/2026

Framework for hydraulic simulation and effective pressure management in water distribution networks

- Enhanced water distribution analysis efficiency by using a Python-integrated EPANET hydraulic modeling framework.
- Designed and tested pressure management strategies (PRVs, pump operations, demand variations) to enhance service reliability.
- Developed Python scripts for automated scenario generation and result analysis in hydraulic simulations.
- Conducted a case study on a medium-scale WDN to evaluate optimization in pressure distribution and leakage reduction.

GIS & Remote Sensing Applications | Coursework Projects

07/2024 - 12/2025

- Processed DEM data in QGIS (merging, sink filling, flow direction, flow accumulation) to generate drainage networks and watershed boundaries.
- Performed land use/land cover (LULC) classification using unsupervised (K-means) and supervised (SVM) methods on Landsat 8/9 satellite imagery.
- Applied remote sensing–based indices on Sentinel-2 imagery to estimate total suspended solids (TSS) and prepared GIS maps for technical reporting.

EDUCATION

M.Tech, Water Engineering and Management

Indian Institute of Technology (IIT) Kharagpur

07/2024 - 05/2026 Kharagpur, West Bengal

CGPA: 8.5 / 10

B.Tech, Agricultural Engineering

Mahatma Gandhi Chitrakoot Gramodaya Vishwavidyalaya

08/2018 - 05/2022 Madhya Pradesh

CGPA: 7.2 / 10

SKILLS

Core Skills – Water and Environmental Engineering

EPANET | WaterGEMS | SewerGEMS | QGIS | HEC-RAS | Basic InfoWorks ICM | AutoCAD | ArcGIS

Supporting Skills – Analytics, Automation & Project Controls

Python | SQL | EDA | Tableau | Microsoft Power BI | Canva | Microsoft Excel | Microsoft Word | Google Sheets | Claude AI

CERTIFICATION

Hydraulic Modeling with HEC-RAS

Offered by UNESCO in collaboration with the USACE Hydrologic Engineering Center

Bentley Accredited Water Modeler

Bentley Systems

KEY ACHIEVEMENTS

GATE Exam Achievement

Secured All India Rank 125 in GATE (AG) 2024

COURSE WORK

Geo Informatics Laboratory | River Basin Planning and Management | Environmental Hydrology & Hydraulics | Water Engineering Lab | Sustainable Engineering and Circular Economy | Planning and Design of Water Engineering Facilities | Wastewater Management | Water Economics & Governance | Climate Impacts on Water Resources | Disaster Management | Sectoral Water Distribution | Geogenic Water Pollution

LANGUAGES

ENGLISH
Proficient

HINDI
Native